

ISE 5405: Optimization I

Lecture 2: Introduction to Linear Optimization

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Bertsimas–Tsitsiklis, Sections 1.1–1.4

- Setup: notation, LPs, feasibility, and optimality.
- Modeling: production, bakery, diet, power, and scheduling.
- Geometry: graphical solution, four outcomes, and corners.
- Canonical forms: general form, standard form, and conversions.
- LPs in disguise: max objectives and absolute values via epigraphs.

Order matters

We model first and formalize second. Standard-form conversions matter when algorithms enter, after we have built models worth converting.

Notation — Matrices

An $m \times n$ matrix is a rectangular array with m rows and n columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

Course convention

The entry a_{ij} is in row i , column j : row index first. Rows correspond to constraints; columns correspond to variables.

$$A = \begin{bmatrix} 3 & -1 & 0 & 2 \\ 1 & 4 & -2 & 0 \\ 0 & 5 & 1 & -3 \end{bmatrix}.$$

For example, $a_{23} = -2$ and $a_{32} = 5$.

The Transpose: Rows Become Columns

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 2 & 0 & 5 \end{bmatrix} \implies A^T = \begin{bmatrix} 3 & 2 \\ 1 & 0 \\ 4 & 5 \end{bmatrix}.$$

- Bertsimas–Tsitsiklis writes A' ; elsewhere A^T is common.
- $(A^T)^T = A$, and size changes from $m \times n$ to $n \times m$.
- A column vector x is $n \times 1$; x^T is $1 \times n$.

Why care?

$c^T x$ and $a_i^T x$ are row times column, hence single numbers.

Vectors: Columns of Numbers = Points = Arrows

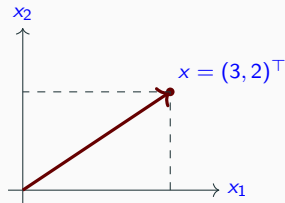
$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n,$$

$$\mathbf{x}^\top = (x_1, x_2, \dots, x_n).$$

Vectors are columns unless transposed. The decision vector $\mathbf{x}$ collects one coordinate per decision.

Three equivalent pictures

$\mathbf{x} = (3, 2)^\top$ is a list of numbers, a point in the plane, and an arrow from the origin. Fluency means switching freely among them.



The Inner Product $x^\top y$ — the Course Workhorse

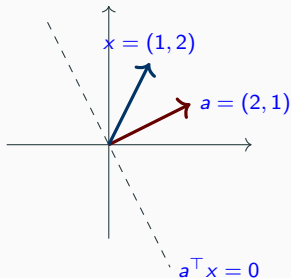
$$x^\top y = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \sum_{i=1}^n x_i y_i.$$

For $x = (1, 3)^\top$ and $y = (2, -1)^\top$,

$$x^\top y = (1)(2) + (3)(-1) = 2 - 3 = -1.$$

Every linear expression is an inner product:

objective $c^\top x$, constraint $a_i^\top x \geq b_i$.



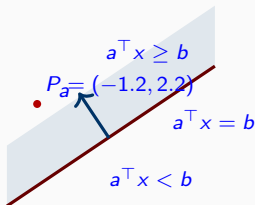
$$a^\top x = (2)(1) + (1)(2) = 4.$$

Sign: positive means acute/same side; negative means obtuse/opposite side; zero means perpendicular.

One Inequality $a^T x \geq b$ = Half the Plane

- $a^T x = b$ is a line in $\mathbb{R}^2$, a plane in $\mathbb{R}^3$, and a hyperplane in higher dimensions.
- a is perpendicular to it and points toward increasing $a^T x$.
- The inequality retains one side: a halfspace.
- Raising b slides the boundary in direction a .

An LP feasible set is the intersection of finitely many halfspaces, which explains flat sides and corners.



$$a = (1.5, 1), \quad b = 1,$$

$$P = (-1.2, 2.2), \quad a^T P = 0.4 < 1.$$

P violates the constraint. Re-aiming a rotates the line; moving P changes the test point; changing b translates the boundary along a .

A Linear Programming Problem (Ex. 1.1)

$$\begin{array}{ll}\min & 2x_1 - x_2 + 4x_3 \\ \text{s.t.} & x_1 + x_2 + x_4 \leq 2, \\ & 3x_2 - x_3 = 5, \\ & x_3 + x_4 \geq 3, \\ & x_1 \geq 0, \quad x_3 \leq 0.\end{array}$$

- Objective $c^\top x$, where $c = (2, -1, 4, 0)^\top$; x_4 has zero objective coefficient.
- Functional rows have one of the forms $a_i^\top x \leq b_i$, $a_i^\top x = b_i$, or $a_i^\top x \geq b_i$.
- Row 1 has $a_1 = (1, 1, 0, 1)^\top$, $b_1 = 2$.
- x_2 and x_4 have no sign restriction: they are free.

The General LP Problem

$$\begin{array}{ll}\min & c^\top x \\ \text{s.t.} & a_i^\top x \geq b_i \quad i \in M_1, \\ & a_i^\top x \leq b_i \quad i \in M_2, \\ & a_i^\top x = b_i \quad i \in M_3, \\ & x_j \geq 0 \quad j \in N_1, \\ & x_j \leq 0 \quad j \in N_2.\end{array}$$

If $j \notin N_1 \cup N_2$, then x_j is free. The index sets sort rows and variables into five ingredient types:

$M_1 : a_i^\top x \geq b_i$	$M_2 : a_i^\top x \leq b_i$	$M_3 : a_i^\top x = b_i$
$N_1 : x_j \geq 0$	$N_2 : x_j \leq 0$	otherwise: free

From $\sum$ Notation to Ax

$$A = \begin{bmatrix} 4 & 5 & 0 \\ 2 & 4 & 1 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad b = \begin{bmatrix} 200 \\ 160 \end{bmatrix}.$$

$$a_1^\top x = 4x_1 + 5x_2 + 0x_3 = \sum_{j=1}^3 a_{1j}x_j,$$

$$a_2^\top x = 2x_1 + 4x_2 + x_3 = \sum_{j=1}^3 a_{2j}x_j,$$

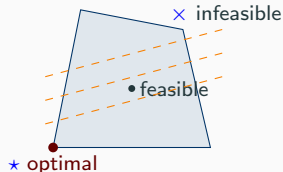
$$Ax = \begin{bmatrix} 4x_1 + 5x_2 \\ 2x_1 + 4x_2 + x_3 \end{bmatrix}.$$

Therefore

$$Ax \geq b \iff \begin{cases} 4x_1 + 5x_2 \geq 200, \\ 2x_1 + 4x_2 + x_3 \geq 160. \end{cases}$$

Basic Definitions

- A **feasible solution** satisfies every constraint.
- The **feasible set** contains all feasible solutions; it may be empty, making the model infeasible.
- The **cost** of feasible x is $c^T x$.
- An **optimal solution** x^* is feasible and satisfies $c^T x^* \leq c^T x$ for every feasible x .

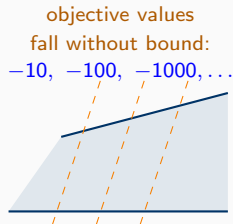


Do not confuse them

Feasibility is a pass/fail test; optimality is winning the entire competition.

Unboundedness & Maximization

If for every K there is a feasible x with $c^T x \leq K$, the optimal cost is $-\infty$; the problem is unbounded.



Maximization and minimization are the same theory:

$$\max_{x \in S} c^T x = -\min_{x \in S} (-c)^T x.$$

The same point wins both contests after reversing the objective vector.

Course convention

We minimize by default and convert a maximization by changing the objective sign.

Why $\min c^T x$ with $Ax \geq b$?

Our convention (following BT)

$$\min c^T x \quad \text{s.t.} \quad Ax \geq b.$$

Natural story: minimize cost while meeting requirements, as in diet or staffing.

Other books

$$\max c^T x \quad \text{s.t.} \quad Ax \leq b.$$

Natural story: maximize profit within resource limits. Sign changes make the two forms equivalent.

Minimizing also matches convex optimization:

$$\underbrace{\min c^T x}_{\text{LP}} \subset \underbrace{\min f(x), f \text{ convex.}}_{\text{convex optimization}}.$$

Local/global results, optimality conditions, and duality then transfer naturally to nonlinear convex settings.

Guided modeling tutorial

The course-text tutorial takes a verbal decision problem through variables, constraints, and an objective.

[https://open-optimization.github.io/
open-optimization-or-book/visualizations](https://open-optimization.github.io/open-optimization-or-book/visualizations)

Modeling routine

Visit the tutorial after class for its guided interaction. In class, name the decisions first; then attach units to every coefficient and right-hand side.

A Production Problem

Produce n goods from m materials. Let b_i be available material i , a_{ij} material i used per unit of good j , and c_j revenue per unit. Define x_j as the amount of good j produced.

$$\begin{aligned} \max \quad & \sum_{j=1}^n c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^n a_{ij} x_j \leq b_i, \quad i = 1, \dots, m, \\ & x_j \geq 0, \quad j = 1, \dots, n. \end{aligned}$$

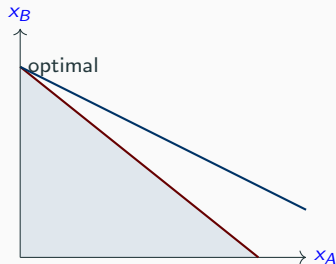
Each row is one material budget; each column is one good's resource profile.

Modeling Sandbox: Production Planning

For the displayed production instance:

$$\begin{array}{ll}\max & 40x_A + 55x_B \\ \text{s.t.} & 4x_A + 5x_B \leq 200, \\ & 2x_A + 4x_B \leq 160, \\ & x_A, x_B \geq 0.\end{array}$$

Corner	Plan	Profit
origin	(0, 0)	\$0
labor intercept	(50, 0)	\$2,000
shared y -intercept	(0, 40)	\$2,200



Widget lesson

The four controls vary the two profits and two capacities. At the displayed values the optimal plan is $x_A = 0, x_B = 40$, profit \$2,200; labor and material both bind, with 0 hours and 0 kg left. Corners can jump at parameter breakpoints.

Bakery Production (3 variables)

	Cake	Cookie	Muffin	Avail.
Profit	\$6.00	\$1.50	\$2.50	—
Flour (g)	500	100	200	4,800
Sugar (g)	200	50	80	2,060
Oven (min)	60	10	20	480

$$\begin{aligned} \max \quad & 6x_1 + 1.5x_2 + 2.5x_3 \\ \text{s.t.} \quad & 500x_1 + 100x_2 + 200x_3 \leq 4800, \\ & 200x_1 + 50x_2 + 80x_3 \leq 2060, \\ & 60x_1 + 10x_2 + 20x_3 \leq 480, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

With three variables the picture leaves the plane, but the recipe is unchanged. This bakery returns when we pivot simplex by hand.

Practice: the Welding Robots

There are 21 units of transparent aluminum alloy, 23 joining hours, and 17 cutting hours. Bookcases, desks, and cabinets sell for \$18, \$16, and \$10. Their (alloy, join, cut) use is respectively (2, 3, 1), (2, 2, 2), and (1, 2, 1).

Your turn — 2 minutes

Name the decision variables; then formulate objective, three resource rows, and domains.

Revealed solution, with x_i the number of units of product i :

$$\begin{array}{ll}\max & 18x_1 + 16x_2 + 10x_3 \\ \text{s.t.} & 2x_1 + 2x_2 + x_3 \leq 21 \quad \text{alloy,} \\ & 3x_1 + 2x_2 + 2x_3 \leq 23 \quad \text{joining,} \\ & x_1 + 2x_2 + x_3 \leq 17 \quad \text{cutting,} \\ & x_1, x_2, x_3 \geq 0.\end{array}$$

Example 1.3 — The Diet Problem

There are n ingredients and m nutrients. Ingredient j costs c_j per unit and contributes column A_j of nutrients; b is the ideal nutrient vector.

$$\begin{aligned} \min \quad & \sum_{j=1}^n c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^n A_j x_j = b, \\ & x_j \geq 0, \quad j = 1, \dots, n. \end{aligned}$$

If b instead gives minimum requirements, replace equality by $Ax \geq b$. This is the classic Stigler diet LP.

Multiperiod Planning of Electric Power

Plan capacity for T years; year- t demand is d_t , existing oil capacity is e_t . Coal lasts 20 years and costs c_t per MW; nuclear lasts 15 years and costs n_t per MW. Nuclear may never exceed 20% of total capacity.

Variables

x_t, y_t : coal/nuclear capacity brought online at the start of year t . w_t, z_t : total coal/nuclear capacity available during year t .

$$\min \sum_{t=1}^T (c_t x_t + n_t y_t).$$

The next frame turns lifetime, demand, and policy statements into constraints.

Electric Power — Constraints

$$w_t = \sum_{s=\max\{1, t-19\}}^t x_s \quad \text{coal lasts 20 years,}$$

$$z_t = \sum_{s=\max\{1, t-14\}}^t y_s \quad \text{nuclear lasts 15 years,}$$

$$w_t + z_t + e_t \geq d_t \quad \text{meet demand.}$$

The nuclear-share ratio is linearizable because the denominator has known sign:

$$\frac{z_t}{w_t + z_t + e_t} \leq 0.2 \quad \Longleftrightarrow \quad 0.2w_t - 0.8z_t + 0.2e_t \geq 0.$$

Modeling lesson

Some nonlinear-looking constraints, including ratios of linear terms with known-sign denominators, are secretly linear.

A Scheduling Problem (Nurses)

Night-shift demand on day j is d_j . Each nurse works five consecutive days. Choose x_j , the number of nurses starting a five-day week on day j . Choosing nurses working on day j would not encode consecutive days.

$$\begin{aligned} \min \quad & \sum_{j=1}^7 x_j \\ \text{s.t.} \quad & x_1 + x_4 + x_5 + x_6 + x_7 \geq d_1, \\ & x_1 + x_2 + x_5 + x_6 + x_7 \geq d_2, \\ & \vdots \\ & x_3 + x_4 + x_5 + x_6 + x_7 \geq d_7, \\ & x_j \geq 0, \quad x_j \in \mathbb{Z}, \quad j = 1, \dots, 7. \end{aligned}$$

Row j counts starts within the preceding five days, cyclically.

...But That's an Integer Program

- $x_j \in \mathbb{Z}$ makes the nurse model an integer linear program (ILP), not an LP.
- Dropping integrality gives the LP relaxation. For a minimization model,

$$Z_{LP} \leq Z_{ILP}.$$

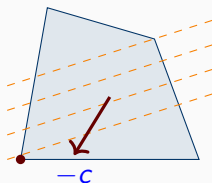
- If an LP optimum is integral, it solves the ILP too.
- Otherwise, rounding up yields a feasible schedule here, not necessarily an optimal one, and the LP bound measures a gap.

Seed of integer programming

Bounds plus rounding lead toward branch-and-bound and ISE 5406.

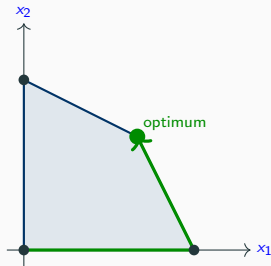
Graphical Representation

- $a_1x_1 + a_2x_2 = b$ is a line dividing $\mathbb{R}^2$ into two halfspaces.
- (a_1, a_2) is orthogonal to the line and points toward increasing left-hand side.
- Objective level sets $c^\top x = t$ are parallel lines orthogonal to c .
- To minimize, slide the line in direction $-c$ while retaining contact with the feasible set.



2D LP Grapher

$$\begin{array}{ll}\min & -x_1 - x_2 \\ \text{s.t.} & x_1 + 2x_2 \leq 3, \\ & 2x_1 + x_2 \leq 3, \quad x_1, x_2 \geq 0.\end{array}$$



corner	$-x_1 - x_2$
(0, 0)	0
(0, 1.5)	-1.5
(1, 1)	-2
(1.5, 0)	-1.5

Status: minimum -2 at $(1, 1)$. A valid simplex walk is

$$(0, 0) \rightarrow (1.5, 0) \rightarrow (1, 1),$$

with objective $0 \rightarrow -1.5 \rightarrow -2$. The next frame separates the possible LP outcomes.

The Four Possibilities

Fix $P = \{x : -x_1 + x_2 \leq 1, x_1, x_2 \geq 0\}$ and vary c :

1. $c = (1, 1)^T$: unique optimum $(0, 0)$.
2. $c = (1, 0)^T$: multiple optima on the bounded edge $\{(0, x_2) : 0 \leq x_2 \leq 1\}$.
3. $c = (0, 1)^T$: multiple optima on an unbounded set, the nonnegative x_1 -axis.
4. $c = (-1, -1)^T$: objective tends to $-\infty$.

Adding $x_1 + x_2 \leq -2$ makes the feasible set empty.

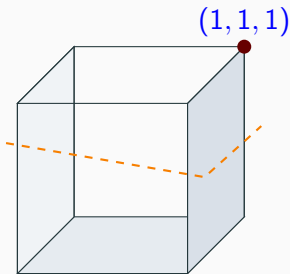
Important distinction

An unbounded feasible set may still have an attained, finite optimum; objective direction decides whether the problem is unbounded.

Example 1.7 in 3D — Spin It Yourself

$$\begin{array}{ll}\min & -x_1 - x_2 - x_3 \\ \text{s.t.} & 0 \leq x_i \leq 1, \quad i = 1, 2, 3.\end{array}$$

The feasible set is the unit cube:
8 corners, 12 edges, and 6 faces.
Objective level sets are planes.
Sliding them lands at $(1, 1, 1)$,
with cost -3 .



Scale

The n -cube has 2^n corners, more
than a googol for $n = 333$.

Higher Dimensions & the Key Observation

- With three variables, level sets of $c^T x$ are planes perpendicular to c ; minimization slides them in direction $-c$.
- In standard form, $Ax = b$ confines x to an $(n - m)$ -dimensional affine set; within it, only $x \geq 0$ constrains us.
- Example: $x_1 + x_2 + x_3 = 1$, $x \geq 0$ is a two-dimensional triangle living in $\mathbb{R}^3$.

Key observation

In every example, if an optimal solution exists, one can be found at a corner of the feasible set. Chapter 2 proves this; Chapter 3 turns it into the simplex algorithm.

A Simpler Form: All Constraints as " $\geq$ "

Every general LP feasible set can use only $a_i^\top x \geq b_i$:

$$a_i^\top x \leq b_i \iff -a_i^\top x \geq -b_i,$$

$$a_i^\top x = b_i \iff \begin{cases} a_i^\top x \geq b_i, \\ -a_i^\top x \geq -b_i. \end{cases}$$

Sign restrictions are already rows of this type using $\pm e_j$. Stack row vectors and right-hand sides:

$$\min c^\top x \quad \text{s.t.} \quad Ax \geq b.$$

This is the general form used for LP theory.

Ex. 1.2 → General Form $Ax \geq b$

Start with

$$\min 2x_1 - x_2 + 4x_3 \quad \text{s.t.} \quad \begin{cases} x_1 + x_2 + x_4 \leq 2, \\ 3x_2 - x_3 = 5, \\ x_3 + x_4 \geq 3, \\ x_1 \geq 0, \quad x_3 \leq 0. \end{cases}$$

The displayed reveal stages give

$$\begin{aligned} -x_1 - x_2 - x_4 &\geq -2, & 3x_2 - x_3 &\geq 5, & -3x_2 + x_3 &\geq -5, \\ x_3 + x_4 &\geq 3, & x_1 &\geq 0, & -x_3 &\geq 0. \end{aligned}$$

Thus

$$A = \begin{bmatrix} -1 & -1 & 0 & -1 \\ 0 & 3 & -1 & 0 \\ 0 & -3 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix}, \quad b = \begin{bmatrix} -2 \\ 5 \\ -5 \\ 3 \\ 0 \\ 0 \end{bmatrix}, \quad c = \begin{bmatrix} 2 \\ -1 \\ 4 \\ 0 \end{bmatrix}.$$

Standard Form

$$\begin{array}{ll}\min & c^T x \\ \text{s.t.} & Ax = b, \\ & x \geq 0.\end{array}$$

Equalities and nonnegative variables only: the form used by simplex.

Column interpretation

$$Ax = \sum_{j=1}^n A_j x_j.$$

Thus $Ax = b$ says to synthesize b from columns A_j , using nonnegative amounts x_j at unit costs c_j .

Reduction to Standard Form

Standard form is a special case of general form, and every LP can be transformed to an equivalent standard-form LP.

General form	Standard form
$\min c^T x$	$\min \tilde{c}^T z$
$a_i^T x$ with $\geq, \leq$, or $=$	$\tilde{A}z = \tilde{b}$
$x_j \geq 0$, $x_j \leq 0$, or free	$z \geq 0$

1. Eliminate nonpositive and free variables.
2. Eliminate inequality constraints.

The payoff is needing algorithms for only one canonical form.

The Four Transformation Rules

1. For $x_j \leq 0$, replace $x_j = -x'_j$ with $x'_j \geq 0$.
2. For free x_j , write $x_j = x_j^+ - x_j^-$, with $x_j^+, x_j^- \geq 0$.
3. For $\sum_j a_{ij}x_j \leq b_i$, add slack:

$$\sum_j a_{ij}x_j + s_i = b_i, \quad s_i \geq 0.$$

4. For $\sum_j a_{ij}x_j \geq b_i$, subtract surplus:

$$\sum_j a_{ij}x_j - s_i = b_i, \quad s_i \geq 0.$$

The conversion adds variables, a cost algorithms gladly pay.

Ex. 1.4 → Standard Form

$$\min 2x_1 + 4x_2 \quad \text{s.t.} \quad x_1 + x_2 \geq 3, \quad 3x_1 + 2x_2 = 14, \quad x_1 \geq 0,$$

where x_2 is free. Set $x_2 = x_2^+ - x_2^-$ and subtract surplus x_3 :

$$\begin{aligned} \min \quad & 2x_1 + 4x_2^+ - 4x_2^- \\ \text{s.t.} \quad & x_1 + x_2^+ - x_2^- - x_3 = 3, \\ & 3x_1 + 2x_2^+ - 2x_2^- = 14, \\ & x_1, x_2^+, x_2^-, x_3 \geq 0. \end{aligned}$$

Two-way mapping examples:

$$(6, -2) \mapsto (6, 0, 2, 1), \quad (8, 1, 6, 0) \mapsto (8, -5).$$

What Does "Equivalent" Mean?

Two minimization problems Π_1, Π_2 are equivalent if both are infeasible or they have the same optimal cost.

Sufficient mapping lemma

They are equivalent if:

- every Π_1 -feasible solution maps to a Π_2 -feasible solution with equal or lower cost; and
- every Π_2 -feasible solution maps to a Π_1 -feasible solution with equal or lower cost.

Use general form $Ax \geq b$ to develop theory; use standard form $Ax = b, x \geq 0$ to run algorithms.

Piecewise Linear Convex Functions

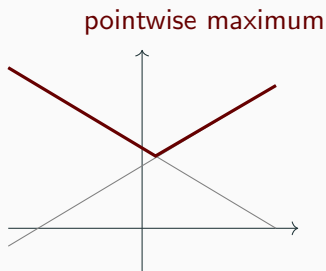
Given $c_i \in \mathbb{R}^n$ and $d_i \in \mathbb{R}$,

$$f(x) = \max_{i=1,\dots,m} (c_i^\top x + d_i)$$

is piecewise linear and convex. The simplest example is

$$|x| = \max\{x, -x\}.$$

Minimizing such a function over a polyhedron is an LP in disguise.



The Epigraph Trick

$$\min_x \max_i (c_i^\top x + d_i) \quad \text{s.t.} \quad Ax \geq b.$$

Introduce z above every affine piece:

$$\begin{aligned} \min_{x,z} \quad & z \\ \text{s.t.} \quad & z \geq c_i^\top x + d_i, \quad i = 1, \dots, m, \\ & Ax \geq b. \end{aligned}$$

At optimality $z^* = \max_i (c_i^\top x^* + d_i)$. Example:

$$\min \max\{2x_1 + 4x_2, 2x_1 + x_2\} \implies z \geq 2x_1 + 4x_2, \quad z \geq 2x_1 + x_2.$$

Absolute Values as an LP

For $c_i \geq 0$,

$$\min \sum_{i=1}^n c_i |x_i| \quad \text{s.t.} \quad Ax \geq b.$$

Use $|x_i| = \min\{z_i : x_i \leq z_i, -x_i \leq z_i\}$:

$$\begin{aligned} \min \quad & \sum_{i=1}^n c_i z_i \\ \text{s.t.} \quad & Ax \geq b, \\ & x_i \leq z_i, \quad -x_i \leq z_i, \quad i = 1, \dots, n. \end{aligned}$$

Warning

Exactness uses minimization and nonnegative coefficients c_i .

For $\min 2|x_1| + x_2$ with $x_1 + x_2 \geq 4$, minimize $2z_1 + x_2$ and add $x_1 \leq z_1, -x_1 \leq z_1$.

Application: Data Fitting

For data (a_i, b_i) , fit $b \approx a^\top x$. Residual magnitude is $|b_i - a_i^\top x|$.

Largest residual (ℓ_∞)

$$\begin{array}{ll}\min & z \\ \text{s.t.} & b_i - a_i^\top x \leq z, \\ & a_i^\top x - b_i \leq z, \quad i = 1, \dots, m.\end{array}$$

Sum of residuals (ℓ_1)

$$\begin{array}{ll}\min & z_1 + \dots + z_m \\ \text{s.t.} & b_i - a_i^\top x \leq z_i, \\ & a_i^\top x - b_i \leq z_i, \quad i = 1, \dots, m.\end{array}$$

Robust regression is LP. Least squares is the ℓ_2 cousin and is not an LP.

Explore More (self-study)

Live demo library

[https:](https://open-optimization.github.io/open-optimization-or-book/visualizations)

[//open-optimization.github.io/open-optimization-or-book/visualizations](https://open-optimization.github.io/open-optimization-or-book/visualizations)

- `#modeling-intro`: words-to-math tutorial.
- `#objective-slider`: objective level-line slider.
- `#concept-quiz`: LP concept questions.
- `#simplex-dictionary`: bakery pivots for Chapter 3.

Reference-deck offline resources

The reference deck also bundled graphical-method, duality, and sensitivity explorers. They remain reference resources, not dependencies of this Fall 2026 PDF.

The course text, *Mathematical Programming and Operations Research*, is available through the same Open Optimization project.

Summary

- LPs come in many shapes; general form $Ax \geq b$ and standard form $Ax = b, x \geq 0$ capture them.
- Free-variable splitting and slack/surplus variables preserve a precise notion of equivalence.
- Production, diet, power, and scheduling show that variable choice is a design decision.
- Piecewise-linear convex objectives, absolute values, and residual fitting reduce to LP through epigraphs.
- Geometry suggests optimal solutions at corners, the idea behind the rest of the course.

Next: proof techniques and LaTeX, then polyhedral geometry.